EE1101: Signals and Systems Short Notes

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June 19, 2023

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1 Signals and Systems

Energy of a continuous time signal (in parentheses discrete time):

$$\boxed{E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt} \qquad \left(E_\infty = \sum_{-\infty}^{\infty} |x[n]|^2\right)$$

Power of a signal:

$$\boxed{P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt} \qquad \left(P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{-N}^{N} |x[n]|^2\right)$$

1.1 Transforming a Signal

Consider the signal $x(t)$. When we mean "transform" we want to plot a new function $y(t)$ **on the same x-axis**, which is a manipulation of the original signal.

a. Shifting horizontally to the left by an amount t_0: $x(t - t_0)$
b. Shifting horizontally to the right by an amount t_0: $x(t + t_0)$
c. Shifting vertically up by an amount x_0: $x(t) + x_0$
d. Shifting vertically down by an amount x_0: $x(t) - x_0$
e. Stretching (making it look fatter) by an amount α: $x(t/\alpha)$
f. Squeezing (making it look thinner) by an amount α: $x(\alpha t)$
g. Shifting horizontally by an amount t_0 **and then** stretching it by an amount α: $x(t/\alpha - t_0)$
h. Stretching it by an amount α **and then** shifting horizontally by an amount t_0: $x(t/\alpha - t_0/\alpha)$

$x(at+b) \rightarrow$ shift by b then scale by a

1.2 Types of Signals

Periodic Signals with a time period T: $x(t) = x(t + nT)$
Even Signals: $x(t) = x(-t)$
Odd Signals: $x(t) = -x(-t)$

conjugate symmetric: $x(t) = x^*(-t)$
conjugate asymmetric: $x(t) = -x^*(-t)$

1.3 Unit Step and Impulse Functions

In discrete time:

$$\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases}$$

The unit step function is the running sum of the unit impulse.

$$u[n] = \sum_{m=-\infty}^{n} \delta[m]$$

$$\boxed{\delta[n] = u[n] - u[n-1]}$$

Sifting property of the δ function: It can extract out the value of the function at n_0 by multiplying by $\delta[n - n_0]$:

$$x[n]\delta[n - n_0] = x[n_0]\delta[n - n_0]$$

In continuous time:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

$$\boxed{\delta(t) = \frac{du(t)}{dt}}$$

$$u(t) = \int_{-\infty}^{t} \delta(\tau) d\tau$$

However the true definition of the δ function:

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$$

for any function $f(t)$.

1.4 Properties of Systems

a. Memoryless: Output is dependent only on the present (**not** even on the future).

b. Invertible: There exists an inverse to the system. In other words, *there is no information loss*.

c. Causal: Output depends on present and past inputs only. Alternatively called *nonanticipatory*.

d. Stable: There exists no bounded input (vertically) which gives us a unbounded output.

$$|x(t)| < B \implies |y(t)| < f(B)$$

e. Time Invariant: A time shift in the input signal corresponds to the exact same time shift in the output.

$$x(t) \to y(t) \implies x(t - t_0) \to y(t - t_0)$$

f. Linear: A linear combination of two inputs should give the same linear combination of the two corresponding outputs.

$$x_1(t) \to y_1(t), x_2(t) \to y_2(t) \implies \alpha_1 x_1(t) + \alpha_2 x_2(t) \to \alpha_1 y_1(t) + \alpha_2 y_2(t)$$

2 Linear Time Invariant Systems

Because the $\delta[n - n_0]$ function sifts the value of the signal at n_0, we can write the signal $x[n]$ as the sum of all the contributions at each n. Thus

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k]$$

The **impulse response** $h[n]$ of a system to be the transformation of the delta function. The convolution sum is defined as:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] = x[n] * h[n]$$

Each term in this sum ($x[k]h[n - k]$) denotes the *contribution* of $x[k]$ to the final output at n.
In continuous time, the convolution integral becomes:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

An LTI system can be completely described by its impulse response. The convolution commutates, associates and distributes:

$$x(t) * h(t) = h(t) * x(t)$$

$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$

$$x(t) * (h_1(t) + h_2(t)) = x(t) * h_1(t) + x(t) * h_2(t)$$

When the input signal is convoluted with the δ function, you get back the same signal.

$$x(t) * \delta(t) = x(t)$$

2.1 Properties of LTI Systems

Examining the behaviour of the impulse response for different types of LTI systems.

a. Memorylessness: $h(t) = 0$ if $t \neq 0$. or $h(t) = A\delta(t)$

b. Causality: $h(t) = 0$ if $t < 0$.

c. Invertibility: Let the inverse impulse response be $h_1(t)$. Then $h(t) * h_1(t) = \delta(t)$.

d. Stability: The impulse response should be absolutely integrable (or absolutely summable).

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty \qquad \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

2.2 Unit Step Response

The response of the system when the input is the unit step function.

$$s(t) = u(t) * h(t) = \int_{-\infty}^{t} h(\tau) d\tau \qquad h(t) = \frac{ds(t)}{dt}$$

In other words the impulse response is the derivative of the step response ($h(t) = s'(t)$).

3 Fourier Series

Consider a **periodic** signal with a time period T. Its fundamental frequency ω_0 is defined as $\omega_0 = 2\pi/T$. Then the Fourier series representation of the signal is given by:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

This is called the **Synthesis equation**. Finding a_k is given by the **Analysis equation**:

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt \qquad a_0 = \frac{1}{T} \int_T x(t) dt$$

3.1 Fourier Series of a Square Wave

Useful to remember this commonly asked Fourier series representation.

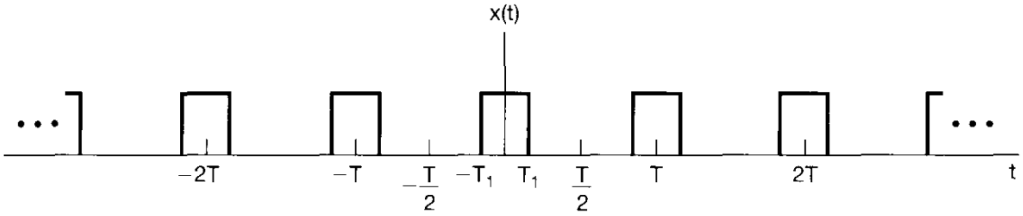


Figure 1: A square wave of duration $2T_1$ and time period T

$$a_0 = \frac{2T_1}{T}$$

$$a_k = \frac{\sin k\omega_0 T_1}{k\pi}$$

Two types of plots which contain information about the nature of the Fourier coefficients:

- The Magnitude Plot: It is a plot of $|a_k|$ versus k.
- The Phase Plot: It is a plot of $\arg a_k$ versus k.

3.2 Convergence of the Fourier Series

Three conditions are imposed on a signal for it to possess a Fourier series representation.

a. It must be absolutely integrable over any period T

$$\int_T |x(t)|dt < \infty$$

b. There should be a finite number of maxima and minima in the signal over any time period.

c. There should be a finite number of discontinuities in any time period and each of these discontinuities should be finite.

Gibbs phenomenon: The Fourier series exhibits high frequency ripples and overshoots near discontinuities.

3.3 Properties of Fourier Series

The Fourier series coefficients of a signal $x(t)$ will be represented by $x(t) \xrightarrow{\mathcal{F}} a_k$. For the sake of the properties listed below consider two signals $x(t) \xrightarrow{\mathcal{F}} a_k$ and $y(t) \xrightarrow{\mathcal{F}} b_k$.

a. Linearity: If $z(t) = \alpha_1 x(t) + \alpha_2 y(t)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = \alpha_1 a_k + \alpha_2 b_k$$

b. Time Shifting: If $z(t) = x(t - t_0)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = a_k e^{-k\omega_0 t_0}$$

c. Time Reversal: If $z(t) = x(-t)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = a_{-k}$$

d. Time Scaling: If $z(t) = x(\alpha t)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = a_k$$

However note that the Fourier series representation has changed due to the change in the fundamental frequency.

e. Multiplication: If $z(t) = x(t)y(t)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l}$$

f. (Period) Convolution: If $z(t) = x(t) * y(t)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = T a_k b_k$$

g. Conjugation: If $z(t) = x^*(t)$

$$z(t) \xrightarrow{\mathcal{F}} c_k = a_{-k}^*$$

h. Differentiation: If $z(t) = \frac{dx(t)}{dt}$

$$z(t) \xrightarrow{\mathcal{F}} c_k = a_k(jk\omega_0)$$

i. Integration: If $z(t) = \int_{-\infty}^{t} x(\tau)d\tau$

$$z(t) \xrightarrow{\mathcal{F}} c_k = \frac{a_k}{jk\omega_0}$$

Special signal properties correspond to special properties that the Fourier coefficients possess.

a. If $x(t)$ is real the real part of the Fourier series coefficients is even and the purely imaginary part is odd. The magnitude plot is even and the phase plot is odd.

b. If $x(t)$ is real and even
$$a_k = a_{-k}^* = a_{-k}$$
which means its Fourier Series coefficients are also real and even.

c. If $x(t)$ is real and odd
$$a_k = a_{-k}^* = -a_{-k}$$
which means its Fourier Series coefficients are **purely imaginary** and odd.

d. If $x(t)$ is purely imaginary and even
$$a_k = -a_{-k}^* = a_{-k}$$
which means its Fourier Series coefficients are purely imaginary and even.

e. If $x(t)$ is purely imaginary and odd
$$a_k = -a_{-k}^* = -a_{-k}$$
which means its Fourier Series coefficients are **real** and odd.

3.4 LTI Systems with Fourier Series

The frequency response of a system with an impulse response $h(t)$ is:

$$\boxed{H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt}$$

If an LTI system has a frequency response $H(j\omega)$ and the input is $x(t) \xrightarrow{\mathcal{F}} a_k$

$$y(t) \xrightarrow{\mathcal{F}} b_k = a_k H(jk\omega_0) \quad \Rightarrow H(j\omega) = \frac{b_k}{a_k}$$

4 The Continuous-Time Fourier Transform

The Fourier Transform of a signal $x(t)$ is:

$$\boxed{X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt}$$

The Inverse Fourier transform is:

$$\boxed{x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega}$$

The Fourier coefficients are given by:

$$\boxed{a_k = \frac{X(j\omega)}{T}|_{\omega=k\omega_0}}$$

The Fourier transform converges under the exact same conditions as specified in 3.2.

4.1 Common Fourier Transform Examples

Consider a square pulse given by $x(t) = \begin{cases} 1 & |t| < T \\ 0 & |t| \geq T \end{cases}$ The Fourier Transform of this is given by:

$$X(j\omega) = \frac{2\sin(\omega T)}{\omega} = 2T sinc\left(\frac{\omega T}{\pi}\right)$$

Consider a periodic function with a Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Then the Fourier Transform of this is given by:

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

The following comprise a list of simple Fourier Transforms:

$$\delta(t - t_0) \rightarrow e^{-j\omega t_0}$$
$$e^{j\omega_0 t} \rightarrow 2\pi\delta(\omega - \omega_0)$$
$$e^{-at}u(t) \rightarrow \frac{1}{a + j\omega}$$

4.2 Properties of Fourier Transforms

Consider two signals $x(t) \xrightarrow{\mathcal{F}} X(j\omega)$ and $y(t) \xrightarrow{\mathcal{F}} Y(j\omega)$. Then the following properties hold:

a. Linearity: $\alpha_1 x(t) + \alpha_2 y(t) \xrightarrow{\mathcal{F}} \alpha_1 X(j\omega) + \alpha_2 Y(j\omega)$

b. Time Shifting: $x(t - t_0) \xrightarrow{\mathcal{F}} X(j\omega)e^{-j\omega t_0}$

c. Conjugation: $x^*(t) \xrightarrow{\mathcal{F}} X^*(-j\omega)$

d. Differentiation in the Time domain: $x'(t) \xrightarrow{\mathcal{F}} j\omega X(j\omega)$

e. Integration: $\int_{-\infty}^{t} x(\tau)d\tau \xrightarrow{\mathcal{F}} \frac{X(j\omega)}{j\omega} + \pi X(0)\delta(\omega)$

f. Time Scaling: $x(at) \xrightarrow{\mathcal{F}} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$

g. Frequency Shifting: $e^{j\omega_0 t}x(t) \xrightarrow{\mathcal{F}} X(j(\omega - \omega_0))$

h. Differentiation in the Frequency domain: $-jtx(t) \xrightarrow{\mathcal{F}} X'(j\omega)$

i. Mapping between Convolution and multiplication: $z(t) = x(t) * y(t) \xrightarrow{\mathcal{F}} Z(j\omega) = X(j\omega)Y(j\omega)$

j. Mapping between Multiplication and convolution:
$z(t) = x(t)y(t) \xrightarrow{\mathcal{F}} Z(j\omega) = \int_{-\infty}^{\infty} X(j\theta)Y(j(\omega - \theta))d\theta$

k. Time Reversal: $x(-t) \xrightarrow{\mathcal{F}} X(-j\omega)$

If $x(t)$ is a real signal then

- $X(j\omega) = X^*(-j\omega)$
- The real part and the magnitude plot of $X(j\omega)$ is even.
- The imaginary part and the phase plot of $X(j\omega)$ is odd.
- If in addition, $x(t)$ were even then $X(j\omega)$ is real and even.
- If in addition, $x(t)$ were odd then $X(j\omega)$ is **purely imaginary** and odd.

4.3 Parseval's Identities

For continuous time periodic signals $x(t) \xrightarrow{\mathcal{F}} a_k$:

$$\boxed{\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2}$$

If $x(t) \xrightarrow{\mathcal{F}} X(j\omega)$ form a Fourier pair then

$$\boxed{\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega}$$

5 The Laplace Transform

The (bilateral) Laplace Transform on a signal $x(t)$ is defined as:

$$\boxed{X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt}$$

where $s = \sigma + j\omega$ is a complex number. The Region of Convergence (ROC) is defined as the range of values of σ for which the Laplace Transform of $x(t)$ (or equivalently the Fourier Transform of $x(t)e^{-\sigma t}$) converges.

5.1 Poles and Zeros

- A pole is defined as the value of s for which $X(s)$ becomes infinite.
- A zero is defined as the value of s for which $X(s)$ becomes zero.

If $X(s)$ is a rational function then it can be represented as $\frac{N(s)}{D(s)}$. If the number of roots of $N(s) = 0$ exceeds the number of roots of $D(s) = 0$ by k then there are k poles at ∞ and if it subceeds then there are k zeros at ∞.

5.2 Region of Convergence of Laplace Transforms

Two definitions before we list the properties of the ROCs: A right-sided signal is one that is zero until a finite time T_1, and a left-sided signal is one that is zero after a finite time T_2. A two-sided signal is one that **is nonzero** until ∞ and $-\infty$.

1. The ROC is always of the form of strips parallel to the $j\omega$-axis in the s-plane.
2. The ROC does not include the poles of a rational Laplace transform and it is bounded only by either the poles or infinity.
3. If $x(t)$ is nonzero for a finite duration and is absolutely integrable then the ROC of its Laplace transform is the entire s-plane.
4. If $x(t)$ is right sided, and if the line $\Re(s) = \sigma_0$ is in the ROC, then all values of s for which $\Re(s) > \sigma_0$ will also be in the ROC. If the Laplace transform is additionally rational then the ROC is the region right of the rightmost pole.
5. If $x(t)$ is left sided, and if the line $\Re(s) = \sigma_0$ is in the ROC, then all values of s for which $\Re(s) < \sigma_0$ will also be in the ROC. If the Laplace transform is additionally rational then the ROC is the region left of the leftmost pole.

5.3 Properties of Laplace Transforms

Consider two signals $x(t) \xrightarrow{\mathcal{L}} X(s)$ with a ROC R_1 and $y(t) \xrightarrow{\mathcal{L}} Y(s)$ with an ROC R_2. Then the following properties hold:

a. Linearity: $\alpha_1 x(t) + \alpha_2 y(t) \xrightarrow{\mathcal{L}} \alpha_1 X(s) + \alpha_2 Y(s)$ with an ROC $R_1 \cap R_2$.

b. Time Shifting: $x(t - t_0) \xrightarrow{\mathcal{L}} X(s)e^{-st_0}$ with the same ROC R_1.

c. Conjugation: $x^*(t) \xrightarrow{\mathcal{L}} X^*(s^*)$ with the same ROC R_1.

d. Differentiation in the Time domain: $x'(t) \xrightarrow{\mathcal{L}} sX(s)$ with the same ROC R_1.

e. Integration: $\int_{-\infty}^{t} x(\tau)d\tau \xrightarrow{\mathcal{L}} \frac{X(s)}{s}$ with an ROC $R_1 \cap \{\Re(s) > 0\}$

f. Time Scaling: $x(at) \xrightarrow{\mathcal{L}} \frac{1}{|a|} X\left(\frac{s}{a}\right)$ with an ROC aR_1.

g. Shifting in the s-domain: $e^{s_0 t}x(t) \xrightarrow{\mathcal{L}} X(s - s_0)$ with an ROC $R_1 + \Re(s_0)$.

h. Differentiation in the Frequency domain: $-tx(t) \xrightarrow{\mathcal{L}} X'(s)$ with the same ROC R_1.

i. Mapping between Convolution and multiplication: $z(t) = x(t) * y(t) \xrightarrow{\mathcal{L}} Z(s) = X(s)Y(s)$ with an ROC $R_1 \cap R_2$.

j. Mapping between Multiplication and convolution:
$z(t) = x(t)y(t) \xrightarrow{\mathcal{L}} Z(j\omega) = \int_{-\infty}^{\infty} X(j\theta)Y(j(\omega - \theta))d\theta$ with an ROC $R_1 \cap R_2$.

k. Time Reversal: $x(-t) \xrightarrow{\mathcal{L}} X(-s)$ with an ROC $-R_1$.

5.4 Some Common Laplace Transforms

Mentioned below are some common Laplace Transforms with their ROCs. To prevent a long list, mentioned below are the most general forms of their type.

$$\delta(t - T) \xrightarrow{\mathcal{L}} e^{-sT} \qquad \Re(s) > -\infty$$

$$u(t) \xrightarrow{\mathcal{L}} \frac{1}{s} \qquad \Re(s) > 0$$

$$-u(-t) \xrightarrow{\mathcal{L}} \frac{1}{s} \qquad \Re(s) < 0$$

$$\frac{t^{n-1}}{(n-1)!}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s^n} \qquad \Re(s) > 0$$

$$e^{-at}u(t) \xrightarrow{\mathcal{L}} \frac{1}{(s+a)} \qquad \Re(s) > -a$$

$$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t) \xrightarrow{\mathcal{L}} \frac{1}{(s+a)^n} \qquad \Re(s) > -a$$

$$-\frac{t^{n-1}}{(n-1)!}e^{-at}u(-t) \xrightarrow{\mathcal{L}} \frac{1}{(s+a)^n} \qquad \Re(s) < -a$$

$$\cos(\omega_0 t)u(t) \xrightarrow{\mathcal{L}} \frac{s}{s^2 + \omega_0^2} \qquad \Re(s) > 0$$

$$\sin(\omega_0 t)u(t) \xrightarrow{\mathcal{L}} \frac{\omega_0}{s^2 + \omega_0^2} \qquad \Re(s) > 0$$

5.5 LTI Systems with Laplace Transform

Note that the Laplace Transform of the impulse response of a system is known as the system function.

a. Causality: For a system with a rational system function, causality of the system is equivalent to the ROC being the right-half plane to the right of the rightmost pole. In general causality implies a ROC containing ∞.

b. Stability: The stability of a system is equivalent to the ROC including the $j\omega$-axis. A causal system with rational system function $H(s)$ is stable if and only if all of the poles of $H(s)$ lie in the left-half of the s-plane-i.e., all of the poles have negative real parts.

5.6 The Unilateral Laplace Transform

The unilateral Laplace Transform is given by:

$$\xi(s) = \int_{0^-}^{\infty} x(t)e^{-st}dt$$

All properties of the unilateral Laplace Transform are identical to the bilateral Laplace transform except for differentiating in the time domain. In that case

$$\frac{dx(t)}{dt} \xrightarrow{\mathcal{L}} s\xi(s) - x(0^-)$$

5.7 Initial and Final Value Theorems

The initial value theorem (in terms of the bilateral Laplace transform $x(t) \xrightarrow{\mathcal{L}} X(s)$) states that given a signal $x(t)$ which is 0 for $t < 0$ and contains no impulses or higher order singularities at the origin then

$$x(0^+) = \lim_{s\to\infty} sX(s)$$

In addition if $x(t)$ has a finite limit at infinity then the final value theorem says that

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$$

Alternatively these statements can be stated in terms of the unilateral Laplace transform.

Then the initial value theorem states that as long as the signal $x(t)$ contains no impulses or higher order singularities at $t = 0$ then

$$x(0^+) = \lim_{s\to\infty} s\xi(s)$$

In addition if $x(t)$ has a finite limit at infinity then the final value theorem says that

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} s\xi(s)$$